Project Proposal: Auditory Stimulation Classification

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https://github.com/fengyiran1005/EEG-Auditory-Stimulation-CNN

Introduction

The human brains exhibit different neural activity patterns sensitive to external stimuli. The project aims to develop a model that would perform a binary classification between EEG of individuals listening to music and individuals in a quiet resting state.

The project is motivated by the significance of EEG signals in various neuroscience research. For example, EEG remains the primary source for epilepsy diagnosis and may also serve as a monitor for patients with neurological disorders, brain tumors, brain injury, etc (1).

Without deep learning, EEG signals were primarily examined visually for manual feature extraction, leading to inaccuracies and misinterpretations due to bias or oversight. This project would implement a 1D Convolutional Neural Network (1D CNN) since the data used was presented in time series data. Specifically, 1D CNN is the most suited for this task since they can perform deep learning directly from raw or minimally processed time-series data, detecting characteristic waveforms, oscillatory patterns, and cross-channel relationships. Moreover, Carvajal-Dossman et al. have also confirmed that a CNN model yielded the best result on public data with a general accuracy above 90% (2).

Background & Related Work

Electrophysiological behavior at auditory stimulation provides meaningful insights into neuronal health. Francavilla et al. discovered that SYNGAP1 haploinsufficiency, a neurological disorder causing neurological disorders like developmental delay, epilepsy, etc, leads to auditory processing deficits (3). Jadhav et al. studied the EEG of SYNGAP1 haploinsufficient mice, confirming that the abnormal oscillations during auditory stimuli are distinguishable with quantitative EEG analysis, demonstrating that neural oscillation patterns serve as biomarkers for auditory processing dysfunction in neurodevelopmental disorders (4). Therefore, a deep learning model in EEG during auditory stimuli would facilitate diagnosis and characterization of such neuronal abnormalities.

Qu et al. used a traditional Random Forest model and Cascade Forward Backpropagation Neural Network (CFBNN) to analyze EEG during neural stimuli to improve disorders of consciousness (DOC) diagnosis. Specifically, they extracted ApEn values from raw EEG data and used a 3 layer CFBNN, which yielded a 83.33% accuracy (5). Aellen et al. adopted the CNN approach to perform a binary prediction of survival and non survival from coma. Specifically, the model used a CNN model with the filter length of the first convolutional layer being 512, then the ReLu activation function, binary cross-entropy loss function and the Adaptive Moment Estimation (Adam) with a learning rate of $5 \times 10\hat{}6$ was applied (6). Vandecappelle et al. have also used the CNN approach to study EEG during auditory stimulation to identify the auditory attention when multiple speakers are present. The model inputted a 64xT EEG matrix to a single convolution layer consisting of 5 spatio-temporal filters producing 5 feature maps and was followed by a ReLu activation function. The model resulted in a binary classification of attention direction, either left or right (7).

Data Processing

The data was obtained from Physionet (9), where 20 volunteers were recruited, and EEG was collected at 200Hz located at 4 channels for 2 minute intervals. Specifically, each participant was recorded for 6 sessions in resting state and 6 sessions in auditory stimulation. The resting state was divided into eyes open and eyes closed and the auditory stimulation was classified by different music genres and

headphone types. Nevertheless, to train this binary classification model, the dataset will simply be classified into resting and stimulation with no dataset discarded to ensure a large and diverse data population to recognize different types of auditory stimuli.

The data provided in 1D time series was segmented and filtered with noise minimized. For this project, each of the 2-minute EEG signals will be z score normalized per channel to preserve characteristics at each electrode. Moreover, the 2-minute interval containing 24000 time points will be segmented into 4 30-second windows with 6000 points to increase the sample count and decrease the training complexity (10). As a result, 240 data sets will be generated respectively for resting and music listening, where 70% of the data will be used for training, 15% for evaluation and 15% for testing.

Architecture

The project would implement a 1D CNN to analyze the EEG time series data. The network consists of 3 main blocks: a) data processing block for normalization, slicing and labels, b) the 3 layer CNN with ReLu activation and max pooling, c) a fully connected layer that generates a 2-unit softmax binary output. Specifically, the input to the CNN block would consist of 4*6000 time points and the 3 layered CNN will have kernels of decreasing size for hierarchical feature learning with the early layer focusing on the slow rhythm and the deeper layers for faster oscillations.

Illustration

A 3 layer CNN will be implemented with full workflow in the following graph: 1.

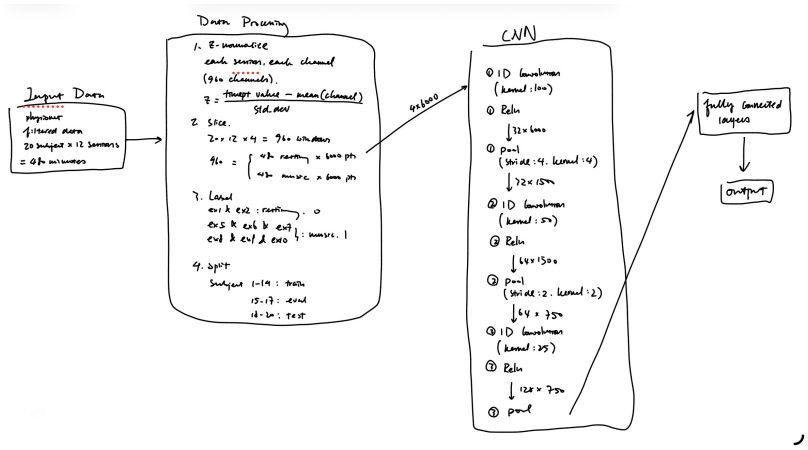

Figure 1: CNN Illustration.

Baseline Model

Inspired by Qu et al. (5), a Random Forest model composed of forests of decision trees examining different features of the dataset will be implemented as the baseline. For the final classification, the majority voting of all trees will be applied. For the auditory stimulation classification, from each of the 30s windows, the random forest model will extract features including frequency of waves (gamma, alpha, beta, etc), statistical values (mean, variance, skewness), and inter-channel relationships.

Ethical Considerations

The data from Physionet was collected with informed consent from all participants with their identity masked. Throughout this project, there will be no attempt to re-identify the participants or misuse the data. More rigorous tests of this model should be performed if being applied in health care to avoid inaccurate predictions. Furthermore, due to the small number of participants, the data might not have a full representation of the population, potentially leading to bias in the algorithm.

References

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